

Tailored Cis–Trans Isomeric Metal–Covalent Organic Frameworks for Coordination Configuration-Dependent Electrochemiluminescence

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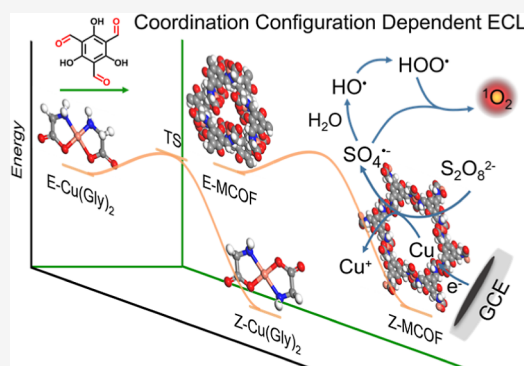


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ABSTRACT: Precise manipulation of the coordination configuration within substances can modulate the band structure and catalytic properties of the target material. Metal–covalent organic frameworks (MCOFs), a crystal material amalgamating the benefits of metal–organic frameworks (MOFs) and covalent organic frameworks (COFs), can integrate a predetermined coordination environment into the frameworks for amplifying the catalytic effect. In this study, we delicately synthesize isomeric MCOFs using bis(glycinato)copper as the aminoligand via kinetically and thermodynamically favorable pathways to yield cis-MCOF and trans-MCOF products, respectively, thereby introducing a cis–trans isomeric coordination field into the framework. Moreover, the twisted skeleton derived from the flexibility of amino acid and β -ketoenamine linkages endows trans-MCOF with surprising water dispersibility. Compared to cis-MCOF, the trans isomerism displays a significant enhancement in cathodic electrochemiluminescence via the catalysis of Cu nodes toward $K_2S_2O_8$. The density of states analysis shows that the d-band center of trans-MCOF is closer to the Fermi level, leading to more stable adsorption binding to promote the catalysis. This study is the first report on constructing predesign coordination configuration MCOFs via an easy-handling method, which gives the guidelines for the design of amino acid-based MCOF materials.



INTRODUCTION

Framework materials, a category of porous polymers, exhibit well-defined periodic structures composed of organic ligands bonded by various linkages.^{1,2} In catalytic applications, it is observed that metals within the framework play vital roles.^{3,4} Regulating the coordination environment of metal nodes allows for fine-tuning of their electronic structures to optimize catalytic performance, thereby achieving efficient control of chemical reactions and improved catalytic efficiency.^{5,6} Meanwhile, the demands of ligand activity and framework crystallinity compel framework materials to choose subtle reversible environments with suitable reaction barriers during the synthesis process.⁷ This implies constraints in the selection of coordination environments. So, isotropic coordination environments remain predominant for metal–organic frameworks (MOFs) and metalized covalent organic frameworks (COFs) catalysts.^{8,9} For example, pillar-layered MOFs¹⁰ and two-dimensional conductive MOFs^{11,12} have been designed to create non-isotropic coordination environments in MOFs. Additionally, salen-based COFs with cis-coordination have been developed to introduce asymmetrically coordinated reactive metal sites into the COF backbone.^{13,14} Therefore, there is a crucial need for further advancements in the introduction of diverse coordination environments.

Emerging metal–covalent organic frameworks (MCOFs) signify a convergence of organometallic and polymer chemistry.^{15,16} MCOFs employ dynamic covalent bonds to connect complexes with prearranged compositions and orientations, seamlessly integrating the rich inorganic properties of the complexes with the ordered porous structure of the framework materials.^{17,18} Within MCOFs, the periodic integration of active metal sites into the extended framework effectively prevents metal site aggregation.¹⁹ The precise location and microenvironment in MCOFs facilitate the comprehension of the relationship between structure and performance in catalysis.^{20–22} Considering MCOF construction, building blocks are required to have both a stable coordination structure and a group that can be used to generate covalent bonds. The metal units in the currently reported MCOFs are limited to Cu-pyrazolate,^{17,23} octahedral Ti^{IV} ,^{16,24} and Ru^{II} tris(2,2'-bipyridine) complexes.¹⁹ Amino acids, which possess multiple coordinating heteroatoms with

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low cost, are undoubtedly one of the candidate ligands to provide such complex coordination environments. However, the high degrees of rotational freedom and complicated interactions of amino acids remain a serious challenge in the design of their framework structures.²⁵ In addition to the method of postmodifying amino acids on the framework,^{26,27} only a few MOFs with twisted and variable spatial structures have been constructed using short peptides as ligands.^{28–31} Since the reactivity of the amino group coordinated to the metal is greatly reduced compared to that of aromatic amines,³² finding a reliable pathway for amino acid complexes to form stable covalent bonds with organic ligands is a great challenge in the synthesis of amino acid-based MCOFs.

As electrochemiluminescence (ECL) emitters, MOFs and COFs have attracted widespread attentions due to their rich functional and structural designability,^{33–35} high loading capacity of active sites,^{36,37} and high activity of nano confinement effects.^{38–40} The excitons essential for the ECL process typically originate from the reaction of the coreactant with the electrode material.^{41–44} Specifically, peroxydisulfate ($\text{K}_2\text{S}_2\text{O}_8$) as a coreactant in cathodic ECL has been documented to undergo an adsorptive decomposition process on the metal nodes.^{45,46} The rate of this catalytic process correlates with the metal species and the coordination environment of the adsorption site.^{47,48} In order to improve the performance of ECL, more exposed reaction active sites, higher reaction activity, and shorter electron transfer distance need to be finely modulated on the electrode material.^{49–51} Compared with three-dimensional structures, planar structures expose more catalytic active sites, shorten ion/electron transport distances, and remove diffusion barriers, effectively improving conductivity and ECL efficiency.⁵² Therefore, designing and synthesizing thin layered two-dimensional framework materials with precise coordination configuration is a promising method for obtaining more efficient ECL emitters.

Herein, we have chosen 2,4,6-trihydroxybenzene-1,3,5-tricarbaldehyde (Tp) and cis–trans bis(glycinato)copper(II) ($\text{Cu}(\text{Gly})_2$) as ligands to obtain the first example of cis–trans isomerized MCOFs named Z-MCOF and E-MCOF via thermodynamic and kinetic reaction pathways (Figure 1), respectively. The successful synthesis of MCOFs and the identification of cis–trans isomerism were confirmed through powder X-ray diffraction (PXRD), X-ray photoelectron spectroscopy (XPS), selected area electron diffraction (SAED), and other characterizations. Owing to the single bond rotation introduced by enol–keto tautomerism and the charges of the metal units, enol-MCOFs with a thickness of only two layers exhibit exceptional dispersibility in the aqueous phase, which can remain dispersed for several days without aggregation. Furthermore, based on the electrochemical catalysis of transition metal nodes toward coreactant, MCOFs used as cathodic ECL electrode-modified material exhibit superior properties to conductive MOF, where Z-MCOF shows twice ECL intensity than E-MCOF. The density of states (DOS) analysis points out that the difference in the electron distribution of Cu within the coordination environment results in the d-band center position being closer to the Fermi energy level, and thus a better adsorption catalytic activity toward the coreactant. To the best of our knowledge, this is the first example of cis–trans isomeric MCOFs, offering a paradigm for precise modulation of the coordination

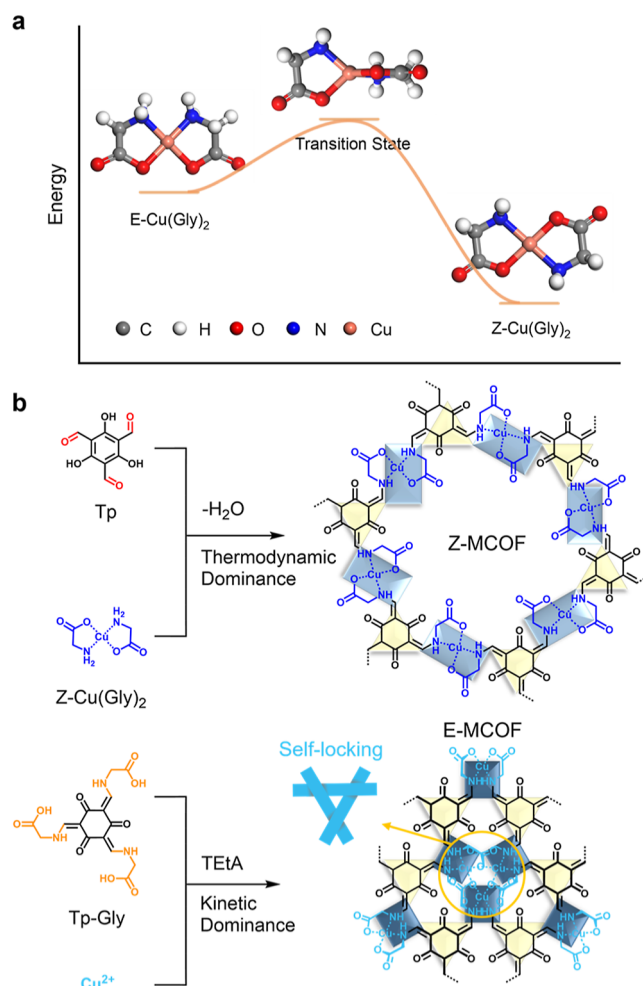


Figure 1. (a) Illustration of the thermodynamic property of $\text{Cu}(\text{Gly})_2$. (b) Synthesis of cis–trans isomeric MCOFs with hcb topology via thermodynamic and kinetic reaction-dominated pathways.

configuration of framework materials for advanced applications.

RESULTS AND DISCUSSION

Design and Synthesis of Cis–Trans Isomeric MCOFs.

By choosing $\text{Cu}(\text{Gly})_2$ as the ligand for linkage, we introduced cis–trans isomerism experimentally into framework materials. As depicted in Figure 1a, cis- $\text{Cu}(\text{Gly})_2$ possesses relatively small activation energy (kinetic advantage) while trans- $\text{Cu}(\text{Gly})_2$ has low reaction enthalpy (thermodynamic advantage), facilitating the cis-to-trans conversion at elevated temperatures.^{53,54} The crystallized $\text{Cu}(\text{Gly})_2$ in a single conformation can be obtained by controlling the reaction temperature (Figure S1a,b). Nevertheless, the complexation of metal ions with amino group results in decreased electron density on nitrogen atoms, thereby diminishing its reactivity and imine bond stability.³² To stabilize imine bonds, Tp, with its enol–keto tautomerism, ensures the formation of a long-range ordered structure through reversible condensation and irreversible proton tautomerism, giving rise to ketone–enamine forms.⁵⁵

Initially, we employed a conventional solvent-thermal reaction approach to copolymerize Tp with both E- $\text{Cu}(\text{Gly})_2$ and Z- $\text{Cu}(\text{Gly})_2$. PXRD characterization demonstrated indistinguishable products for both $\text{Cu}(\text{Gly})_2$ isomers, alongside a

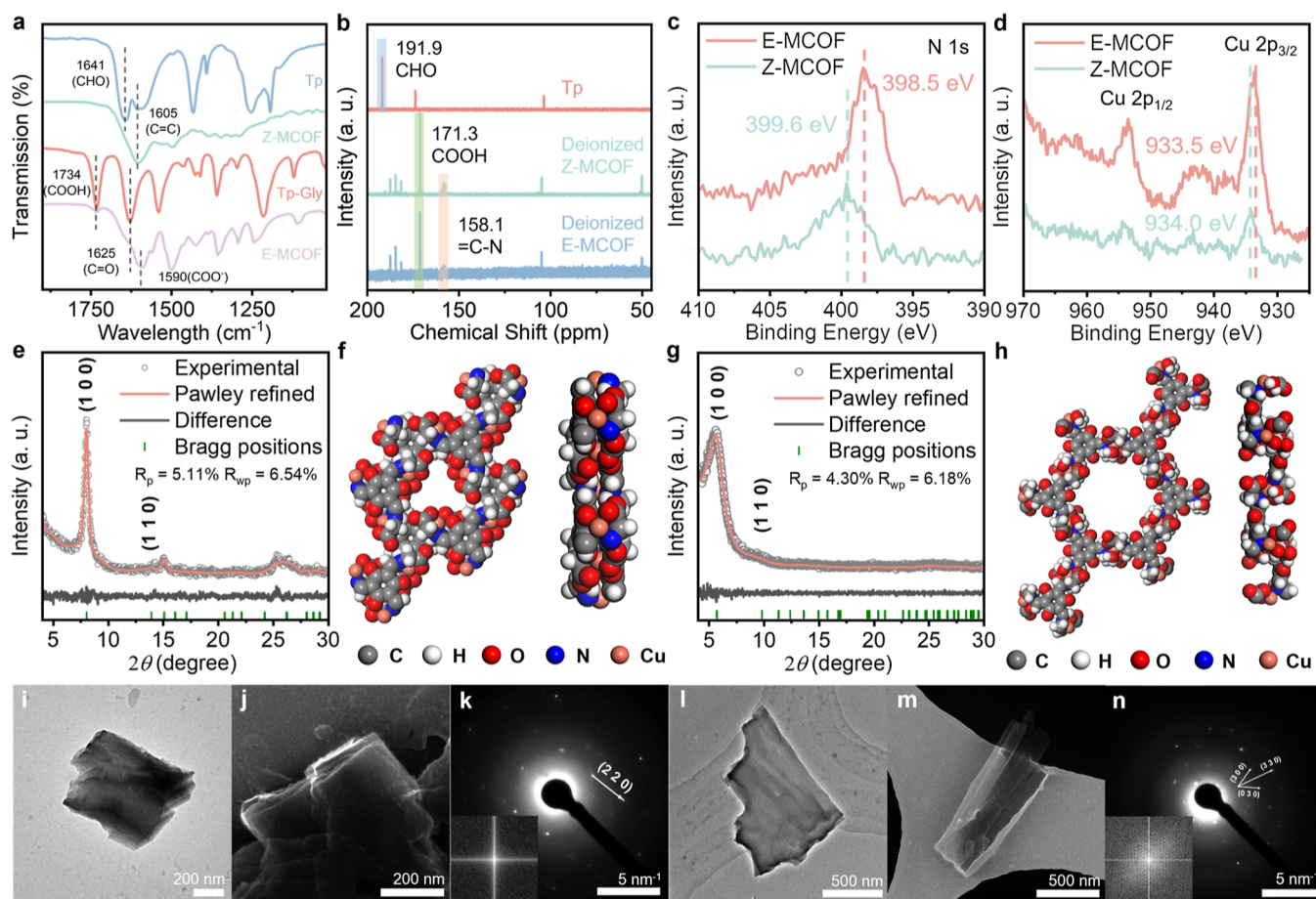


Figure 2. (a) IR spectra of Tp, Tp-Gly, and MCOFs. (b) ^{13}C NMR spectra of Tp and deionized MCOFs. XPS spectra of MCOFs in the (c) N 1s and (d) Cu 2p regions. PXRD patterns of (e) E-MCOF and (g) Z-MCOF with the experimental profiles in gray, the Pawley refined pattern in red, the difference plot in black, and observed reflections in green. Simulated crystal structures of (f) E-MCOF and (h) Z-MCOF for the top (left) and side (right) view. TEM images of the (i) E-MCOF and (l) Z-MCOF. SEM images of (j) E-MCOF and (m) Z-MCOF. SAED patterns of (k) E-MCOF and (n) Z-MCOF. Insets of k and n: lattice diagrams obtained by fast Fourier transform.

significant presence of unreacted E-Cu(Gly)₂ (Figure S1c). Despite the solubility of Cu(Gly)₂ exclusively in aqueous solutions, the coordination environment can undergo inversion in the solvent-thermal process, facilitating the polymerization of only the trans structure with Tp. Given the thermodynamic characteristics of Cu(Gly)₂, we propose a dual-pathway synthesis for cis–trans coordination isomeric MCOFs: one involving thermodynamics (imine bond formation) and the kinetics (coordination bond formation) dominances, as depicted in Figure 1b.

In the thermodynamic pathway, despite extensive optimization of various solvent-thermal conditions, achieving highly crystalline MCOFs posed a significant challenge (Figure S1d). This was mainly attributed to the low solubility of Cu(Gly)₂ in the organic phase. After the reaction with Tp, it was necessary to remove it from the solid surface to proceed with the reaction, leading to reduced reactivity and structural order. To enhance crystallinity, we first obtained a partial reaction precursor of MCOFs through solid grinding in a mortar (Figure S2a). Subsequently, we introduced a small amount of water into the organic phase to regularize the MCOFs. This approach significantly improved the crystallinity in comparison to the solvent-thermal method for trans coordination MCOFs, denoted as Z-MCOF.

In the kinetic pathway, the terminal glycine groups involved in coordination rapidly formed polymers with a cis-configuration at lower temperatures. Tp was sized equivalently to a single benzene ring, as the aldehyde-based ligand. This forces cis-coordinated nodes to overlap in a twisted manner within the channels during the formation of periodic structures, leading to a self-locking structure (Figure 1b). The significant steric hindrance nearly blocks the cis-to-trans conversion, enabling the formation of the thermodynamically disfavored cis coordination configuration product (E-MCOF). In addition, the capacity of these resulting polymers to establish regular frameworks depended on the quantity of triethylamine added (Figure S2b).

Characterization of MCOFs. The chemical structure of MCOFs underwent comprehensive characterization through various techniques, including Fourier transform infrared (FTIR), nuclear magnetic resonance (NMR) spectroscopy, ultraviolet and visible (UV–vis) spectra, energy dispersive spectrometry (EDS), and thermal gravimetric analysis (TGA). Figures 2a and S3 illustrate that the vibrational absorption bands of amino and formyl groups at 3262 and 1641 cm^{-1} disappeared, while the peak around 1610 cm^{-1} broadened toward higher wavenumber, indicating the formation of imine bonds within the framework. The attenuation of the vibration of the carboxylic acid located at 1734 cm^{-1} in Tp–Gly and the

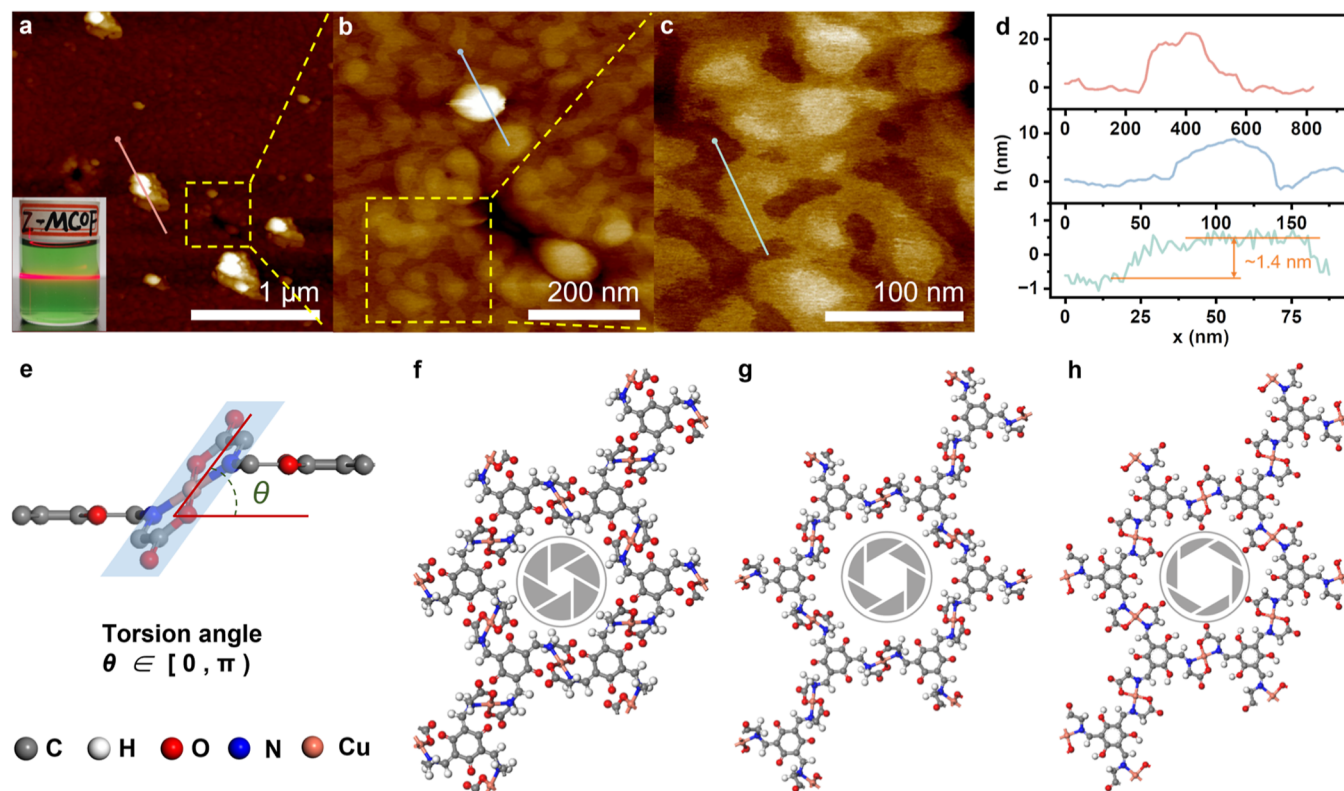


Figure 3. (a) AFM image of Z-MCOF. Inset: photograph of Z-MCOF dispersed in water irradiated with a 650 nm laser. (b,c) Enlargement of the boxed areas. (d) Height distribution of the underlined portions in (a) (red), (b) (blue), and (c) (green). (e) Schematic modeling of torsion due to bis(glycinato)copper in Z-MCOF. Simulated crystal structures of Z-MCOF with torsion angles of (f) 105, (g) 60, and (h) 0°.

shift of the antisymmetric vibration of the COO^- from 1627 to 1590 cm^{-1} in E-MCOF proved the coordination of Tp–Gly to Cu^{2+} . To further illustrate that Tp and $\text{Cu}(\text{Gly})_2$ were connected, we performed liquid NMR characterization of MCOFs after dissociation to remove paramagnetic Cu^{2+} . As shown in Figure 2b, the signal of 191.9 ppm out of the aldehyde group in Tp disappeared, and the appearance of $\text{C}=\text{N}$ connection and carboxylate signals at 158.1 and 171.3 ppm proved the successful formation of MCOFs. UV–visible absorption spectra, as presented in Figure S4, showed that the samples exhibited ~ 650 nm Cu^{2+} d–d transition absorption and strong Tp absorption in the ultraviolet region. The band gaps of E-MCOF and Z-MCOF calculated from ultraviolet diffuse reflectance spectroscopy are 2.28 and 2.09 eV, respectively (Figure S5). EDS confirmed the even distribution of Cu, C, N, and O within the sample volume (Figures S6 and S7), and TGA in an air atmosphere provided that specific Cu content in MCOFs was consistent with the theoretical value of 20% (Figure S8).

The cis–trans coordination isomeric structure of MCOFs was confirmed by using Raman spectroscopy and XPS techniques. Raman spectroscopy was employed to investigate the vibrational absorption modes of Cu coordination nodes at room temperature. In Figure S9, observed Raman features at 330 and 307 cm^{-1} for E- $\text{Cu}(\text{Gly})_2$ and Z- $\text{Cu}(\text{Gly})_2$, were contributed to the whole node oscillation modes in cis- and trans-coordination, respectively. And the Raman adsorption at 560 cm^{-1} was attributed to the $\nu_{\text{Cu}-\text{N}}$ stretching modes coupled to the N–H stretch in cis-coordination.⁵⁶ In MCOFs, hydrogen on N was replaced by carbon, leading to the disappearance of the Raman absorption at 560 cm^{-1} and its

replacement by C–C stretch coupling absorption in the range of 500–580 cm^{-1} . Moreover, the vibrational modes of the MCOF fragments demonstrated the signals at 161, 208, and 279 cm^{-1} for E-MCOF, and 196, 289, and 324 cm^{-1} for Z-MCOF (Figure S10), which matched with the cis-coordination and trans-coordination environment, respectively. XPS furnished information about the atomic environment and concentration on the sample's surface (Table S1). $\text{Cu}(\text{Gly})_2$ exhibited a 0.3–0.6 eV difference in the binding energy positions of N, O, and Cu elements, with the binding energy of cis coordination in a lower field, corresponding with the higher conformational energy (Figures 2c,d, S11). This shift was similarly observed in MCOFs, with the trend aligning with that of $\text{Cu}(\text{Gly})_2$. Following the deconvolution of element peaks (Figure S12), Cu in MCOFs was observed to exist solely in a divalent form. In comparison to $\text{Cu}(\text{Gly})_2$ molecules, the C spectrum in MCOFs revealed the presence of $\text{C}=\text{C}$, and the N spectrum exhibited $\text{C}=\text{N}$, indicating the successful connection of the metal complex with Tp.

The crystallinity of MCOFs with honeycomb (hcb) topologies was assessed using PXRD. E-MCOF exhibited a strong and narrow diffraction peak at 8.04° (Figure 2e), while Z-MCOF displayed a relatively strong but broadened diffraction peak at 5.61° (Figure 2g), suggesting weaker crystallinity in Z-MCOF. The unit cell structures for both MCOF types were constructed using Materials Studio software. E-MCOF adopted the P321 space group with diffraction peaks at 8.02, 13.90, and 15.07° corresponding to (100), (110), and (001) planes (Figure 2f). Z-MCOF adopted the P3 space group with diffraction peaks at 5.68 and 9.78°, corresponding to (100) and (110) planes (Figure 2h). Among

the two planar stacking modes, AA stacking provided a better match with experimental data than AB stacking mode (Figure S13). Pawley refinement results confirmed a good match between the simulated and observed PXRD patterns, revealing specific unit cell parameters (Tables S2,S3)

$$\begin{aligned}\text{E-MCOF: } a &= b = 12.749 \text{ \AA}, c = 5.882 \\ &\text{\AA}, \alpha = \beta = 90^\circ, \gamma = 120^\circ, \\ R_p &= 5.11\% \quad R_{wp} = 6.54\%\end{aligned}$$

$$\begin{aligned}\text{Z-MCOF: } a &= b = 18.269 \text{ \AA}, c = 7.217 \\ &\text{\AA}, \alpha = \beta = 90^\circ, \gamma = 120^\circ, \quad R_p \\ &= 4.30\% \quad R_{wp} = 6.18\%\end{aligned}$$

The morphology and structure of MCOFs were characterized by using transmission electron microscopy (TEM), scanning electron microscopy (SEM), and SAED. E-MCOF exhibited plate-like structures with sizes ranging from 500 nm to 1 μm (Figures 2i,j, S14). Cross-sectional observations revealed a thin plate thickness and a less densely stacked structure. SAED showed limited diffraction bands (Figures 2k, S15) with interplanar spacings of 6.33 and 3.17 $\text{\AA}$, corresponding to the (110) and (220) planes. Meanwhile, Z-MCOF displayed varying morphologies in different solvents: when dispersed in an aqueous phase and dried, it exhibited a smooth, membrane-like surface (Figure S16). In dioxane, it appeared as a stacked blocky material, as shown in Figure 2l,m. SAED results displayed a uniform triangular diffraction pattern (Figure 2n) with an interplanar spacing of 5.27 $\text{\AA}$, corresponding to the (300) plane. N_2 adsorption–desorption experiments of MCOFs conducted at 77 K reveal characteristics of type-I adsorption isotherms (Figure S17). E-MCOF exhibited no adsorption pores for N_2 , whereas Z-MCOF had the most available pore size of 1.25 nm, which is consistent with the simulated structure. To further verify the microporous nature of MCOFs, CO_2 adsorption–desorption tests were performed at 273 K (Figure S18). The adsorption pore size of E-MCOF for CO_2 is 0.45 nm, consistent with the theoretical value of AA-stacked E-MCOF, whereas Z-MCOF has a diverse and broad pore size distribution. The specificity of Z-MCOF may come from the exfoliation due to mechanical milling, which is hampered in the formation of long-range pores and produces chaotic stacking.^{57,58}

Torsional Structure in MCOFs. Z-MCOF exhibits a phenomenon of self-dispersion in water, as illustrated in Figure 3a. Upon introduction of Z-MCOF into the water, rapid dispersion is observed, resulting in a transparent and uniform colloidal solution that remains stably dispersed for several days. Atomic force microscopy (AFM) was employed to characterize the dispersion of the Z-MCOF in water. As shown in Figure 3b–d, the nanosheets of Z-MCOF have diameters ranging from 50 to 200 nm and thicknesses of 1.4 nm with the fluctuation on the exposed plane of 7 $\text{\AA}$, which is close to the single-layer thickness of the simulated structure (Figure S19). The evidence suggests that Z-MCOF can be dispersed into nanosheets with varying numbers of layers, ranging from 2 to 7 layers. Structurally, the source of Z-MCOF's good dispersion in water is the twisted $\text{Cu}(\text{Gly})_2$ in its structure, which enlarges the interlayer spacing and weakens the interlayer π – π stacking interactions. Additionally, the copper's z-axis coordination sites readily interact with water, leading to the solvation of Z-

MCOF, allowing it to exist in the form of a few-layered structure in the aqueous phase.

Turning the view to the Z-MCOF powder, we found that the irregular material exhibits a broad diffraction peak from 5.43° to 6.41° as well as a characteristic peak at 14.56° in the (001) plane of the two-dimensional material (Figure S20a). Varying the amount of water in the reaction mixture revealed a gradual merging of the broad peaks, which may be composed of the (100) plane diffraction of two near-size crystalline phases (Figure S20b). By constructing models, we discovered that the source of the double peaks is due to the flexible, rotatable single bond introduced by Tp. Following the keto-enol transformations, the $\text{Cu}(\text{Gly})_2$ node can be twisted in the xy plane (Figure 3e). Matching the experimental data to optimize the model revealed that the double peaks in PXRD corresponded to two torsion angles of 105° (Figure 3f) and 60° (Figure 3g), denoted as Z-MCOF₁₀₅ and Z-MCOF₆₀, respectively. SAED also captured diffraction patterns in the (300)/(030) planes with crystal plane spacing of 5.58 $\text{\AA}$ in Z-MCOF₆₀ (Figure S21a), as well as in the (220) and (330) planes with crystal plane spacing of 4.09 and 2.68 $\text{\AA}$ in Z-MCOF₁₀₅ (Figure S21b,c). Additionally, we synthesized Z-MCOF in the planar configuration, denoted as Z-MCOF₀ (Figure 3h). It had broad (100) and (001) plane diffraction peaks at 4.74° and 25° and exhibited a broader ultraviolet–visible absorption spectrum due to its better planarity (Figure S22).

It is worth noting that E-MCOF did not exhibit remarkable isomerization or water dispersion characteristics. As shown in Figure S23, E-MCOF is still aggregated as nanoparticles in water, and the root-mean-square roughness is 27.2 nm, much higher than the 1.87 nm of Z-MCOF. Although E-MCOF has a larger interlayer spacing than Z-MCOF₀, it does not exhibit exaggerated (001) plane broadening. We believe that the relatively disordered z-axis stacking has a significant contribution to water dispersibility, which is due to the repulsion between metal node charges. E-MCOF contains the twisted $\text{Cu}(\text{Gly})_2$ within its pores, which restricts both imine bonds to the same side and lacks direct metal node interactions between layers (Figure S24). And the pores provide sufficient space to accommodate adsorbed water molecules, avoiding further weakening of z-axis stacking after water solvation. Moreover, we observed lattice shrinkage and X-ray diffraction peak slip due to small torsion angle changes in the E-MCOF when the reaction temperature was changed (Figure S25). In a word, the twisted structures make MCOFs sufficiently expose Cu to the pores, which favors the distribution of reactive sites within the bulk phase and improves the catalytic performance.

Coordination Configuration-Dependent ECL. In ECL systems, Cu ions have been reported to enhance the ECL intensity by catalyzing the decomposition of $\text{K}_2\text{S}_2\text{O}_8$.^{59–61} There are significant differences in the ECL intensity for copper complex molecules with different coordination environments (Figure S26a). Under equimolar amounts of Cu, Z-MCOF exhibited relatively higher ECL efficiency (98.7% vs 50 μM $[\text{Ru}(\text{bpy})_3]^{2+}/\text{K}_2\text{S}_2\text{O}_8$) than $\text{Cu}(\text{Gly})_2$ (Figure S26b). To compare the ECL properties of different coordination fields in the framework, we additionally synthesized two-dimensional conducting MOFs, Cu-HITP, and Cu-HHTP with CuN_4 and CuO_4 coordination environments (Figures S27 and S28). As shown in Figure 4a, the ECL intensities of all materials increase with the negative potential scanning, and the ECL intensity of Z-MCOF is about twice as high as that of E-MCOF and 20

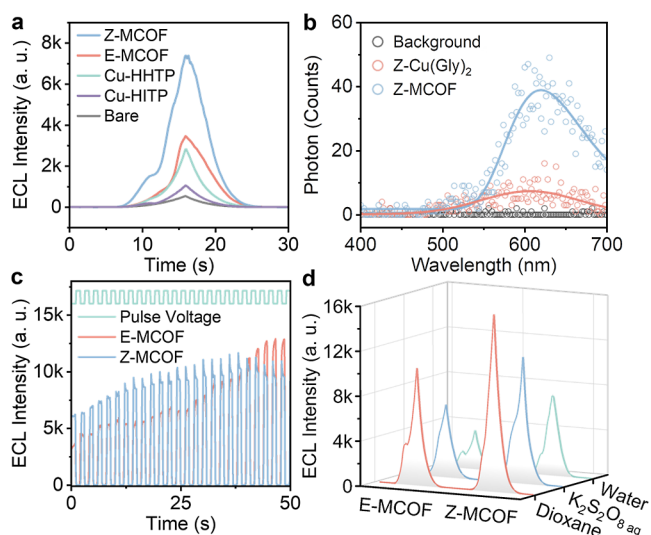


Figure 4. (a) ECL curves of different frameworks modified GCEs from 0 to -1.7 V in 0.1 M PBS (pH = 7.4) containing 0.1 M K₂S₂O₈. Scan rate: 0.1 V s⁻¹. (b) ECL spectra of Z-MCOF and Z-Cu(Gly)₂ collected at -1.7 V for 60 s in 0.1 M K₂S₂O₈. (c) ECL transients of MCOFs by step pulse method from -1.7 to $+1$ V (1 s) in 0.1 M PBS containing 0.1 M K₂S₂O₈ (PMT = 700 V). (d) Effects of ECL intensities at MCOFs modified GCEs on dispersion solvents.

times higher than that of the glassy carbon electrode (GCE). Compared to planar Cu-HHTP, the twisted MCOFs allow the Cu within the bulk to be exposed to the pores for enhanced ECL.

Moreover, the ECL spectrum and EPR tests with 2,2,6,6-tetramethyl-4-piperidone (TEMP) as a scavenger (Figures 4b, S29) confirm that the luminescence originates from the cooperative transition of the singlet oxygen in the $^1\Delta_g$ state rather than from the framework itself.⁶² The deoxygenation ECL experiment shows that the source of singlet oxygen does not depend on dissolved oxygen (Figure S30) but the accumulation of triplet oxygen with the production of singlet

oxygen makes the ECL of MCOFs self-enhancement (Figure 4c).⁶³ When dioxane was used as a dispersant for modifying electrodes, the ECL intensity of Z-MCOF became obviously large compared with water as a dispersant (Figure 4d). And cysteines that tend to bind copper⁶⁴ exhibited a concentration-dependent inhibition of the ECL intensity of Z-MCOF in the range of 100 nM to 1.0 mM (Figure S31). Therefore, we proposed that water molecules occupy the axial coordination vacancy of Cu nodes, making it more difficult for K₂S₂O₈ to adsorb onto copper, thus reducing the catalytic activity for ECL emission.

In addition, comparing the apparent reaction order between ECL and K₂S₂O₈ concentration at the bare and modified electrodes (Figure 5a), it is obvious that the involvement of the copper node changes the reaction path, in which MCOFs exhibit no diffusion-controlled features like bare GCE. The cyclic voltammetric curves illustrate the two-step redox reactions of Cu²⁺/Cu⁺ and Cu⁺/Cu at around -0.3 and -1.1 V, respectively (Figure S32). The K₂S₂O₈ could be electrochemically catalyzed by the reduced Cu to produce sulfate radical anion (SO₄^{•-}) through heteroclinic cleavage. This was demonstrated in EPR tests using 5,5-dimethyl-1-pyrroline N-oxide (DMPO) as a scavenger (Figure 5b), where energizing the Z-MCOF-modified GCE resulted in a significant increase in the amount of SO₄^{•-} and OH[•] in the electrolyte near the electrode,⁶⁵ whereas no radicals for the bare electrode.

To illustrate the influence of the coordination environment, we investigated the DOS changes for the four frameworks (Figures 5c, S33). As a basic understanding, the bonding and strength between a metal and an adsorbate are determined by the position of the d-band center.⁶⁶ As shown in Figure S34, the Cu–N ($L_{\text{Cu-N}} = 1.97$ Å) and Cu–O lengths ($L_{\text{Cu-O}} = 1.89$ Å) in Z-MCOF were shorter than those in E-MCOF (2.08 and 1.92 Å), indicating a tighter binding of the ligands to Cu in Z-MCOF. Figure S35 demonstrates that the d orbitals of Cu atoms in Z-MCOF were more sensitive to the p orbitals of O and N atoms relative to E-MCOF. Thus, the d-band center of E-MCOF was further away from the Fermi level compared to

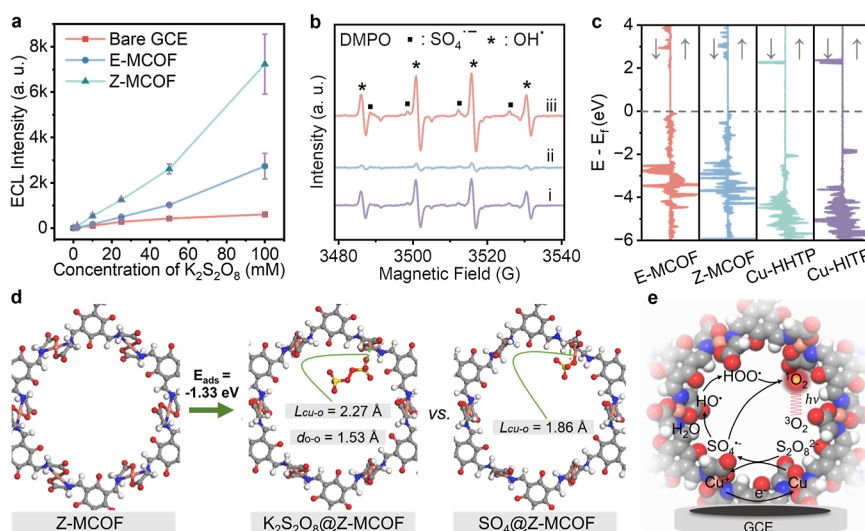


Figure 5. (a) Relationships between ECL intensities at -1.7 V and concentrations of K₂S₂O₈ at different modified GCEs. (b) DMPO-trapped EPR spectra before (i) and (ii,iii) after applying a potential of -1.7 V in 0.1 M K₂S₂O₈ at (ii) GCE and (iii) Z-MCOF-modified GCE. (c) PDOS of the Cu d orbitals for four frameworks. The arrow points out the direction of electron spin. (d) DFT calculations of the adsorbed K₂S₂O₈ molecule catalyzed on the Z-MCOF surface. (e) Schematic diagram of the mechanism of ECL generation catalyzed by MCOFs.

Z-MCOF, rationalizing that Z-MCOF has better catalytic performance. Further, the adsorption process was modeled, and the $K_2S_2O_8$ was not favored to be adsorbed in Z-MCOF bulk with $K_2S_2O_8$ moving away from the catalyst surface after geometry optimization (Figure S36). From the DFT calculations of $K_2S_2O_8$ adsorbed on the Z-MCOF surface, we found that the peroxide bond was activated after adsorption with a long O–O distance ($d_{O-O} = 1.53 \text{ \AA}$), shown in Figure Sd. The adsorption energy was calculated to be -1.33 eV (Table S4). Meanwhile, there were no reactive sites available for $K_2S_2O_8$ adsorption on the surface of E-MCOF without defects (Figure S37). This demonstrates the effectiveness of the strategy of modulating the reaction site activity by controlling the coordination environment. Combining the above analyses and reported ECL mechanisms, the coordination configuration-dependent ECL mechanism is summarized in Figure 5e. MCOFs obtain electrons from the electrode and adsorb $K_2S_2O_8$ on the metal node, and then the $K_2S_2O_8$ is reduced to release sulfate radicals through heterocleavage. The latter attacks water to produce a series of reactive oxygen species, which ultimately generate a singlet oxygen luminescence with the hydroperoxyl.^{60,62} This mechanism explains the phenomenon of ECL enhancement by metal ions for the use of $K_2S_2O_8$ as a coreactant, providing directions for clarifying the fundamentals of such ECL systems.

CONCLUSIONS

In summary, we have successfully demonstrated the feasibility of introducing cis–trans $Cu(Gly)_2$ as a ligand into MCOFs, expanding the boundary of MCOFs composition to amino acid complexes. During the synthesis, we employed both kinetic and thermodynamic reaction pathways to control its coordination mode within the framework, marking the first report of framework materials with cis–trans coordination isomerism. The flexible framework of MCOFs has led to diverse structural variations and exceptional aqueous self-dispersion properties. Notably, the trans coordination structure demonstrates enhanced cathodic ECL, which is derived from the higher d-band centers of Cu resulting from the coordination environment with better adsorption catalysis toward coreactants. This study provides a paradigm for precise control of the coordination configuration of framework materials and expands the potential applications of amino acid-based MCOFs in the catalytic ECL fields.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c02015>.

Materials and methods, detailed synthesis, PXRD, FTIR, NMR, UV–vis, UV-DRS, TEM, AFM, isotherm characterization of the obtained $Cu(Gly)_2$ and MCOFs, ECL tests, EPR, DOS, and adsorption mode calculations, and fractional atomic coordinates for unit cells of MCOFs (PDF)

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Notes

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